

Biomechanics of the knee during closed kinetic chain and open kinetic chain exercises

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ABSTRACT

ESCAMILLA, R. F., G. S. FLEISIG, N. ZHENG, S. W. BARRENTINE, K. E. WILK, and J. R. ANDREWS. Biomechanics of the knee during closed kinetic chain and open kinetic chain exercises. *Med.Sci. Sports Exerc.*, Vol.30, No. 4, pp. 556–569, 1998. **Purpose:** Although closed (CKCE) and open (OKCE) kinetic chain exercises are used in athletic training and clinical environments, few studies have compared knee joint biomechanics while these exercises are performed dynamically. The purpose of this study was to quantify knee forces and muscle activity in CKCE (squat and leg press) and OKCE (knee extension). **Methods:** Ten male subjects performed three repetitions of each exercise at their 12-repetition maximum. Kinematic, kinetic, and electromyographic data were calculated using video cameras (60 Hz), force transducers (960 Hz), and EMG (960 Hz). Mathematical muscle modeling and optimization techniques were employed to estimate internal muscle forces. **Results:** Overall, the squat generated approximately twice as much hamstring activity as the leg press and knee extensions. Quadriceps muscle activity was greatest in CKCE when the knee was near full flexion and in OKCE when the knee was near full extension. OKCE produced more rectus femoris activity while CKCE produced more vasti muscle activity. Tibiofemoral compressive force was greatest in CKCE near full flexion and in OKCE near full extension. Peak tension in the posterior cruciate ligament was approximately twice as great in CKCE, and increased with knee flexion. Tension in the anterior cruciate ligament was present only in OKCE, and occurred near full extension. Patellofemoral compressive force was greatest in CKCE near full flexion and in the mid-range of the knee extending phase in OKCE. **Conclusion:** An understanding of these results can help in choosing appropriate exercises for rehabilitation and training. **Key Words:** CLOSED KINETIC CHAIN, OPEN KINETIC CHAIN, MUSCLE ACTIVITY, PCL, ACL, PATELLOFEMORAL, TIBIOFEMORAL, JOINT FORCE

In 1955, Steindler (54) defined two types of exercises: closed kinetic chain exercises (CKCE) and open kinetic chain exercises (OKCE). In a CKCE, the terminal or distal segment is opposed by “considerable resistance”; in a OKCE, the distal segment is free to move without any external resistance. If the external resistance is fixed from moving, the system is “strictly and absolutely closed.” These categories are often found to be inaccurate or confusing (44). To reduce confusion, Dillman et al. (16) proposed three categories of exercises: a fixed boundary condition with an external load (e.g., leg press where seat slides and the foot plate is fixed), a movable boundary with an external load (e.g., leg press where the seat is fixed and the foot plate moves), and a movable boundary with no external load. In this study CKCE of the leg are defined as exercises in which the feet are fixed from moving and OKCE of the leg are those with no external resistance for movement of the feet.

CKCE—such as squat, leg press, deadlift, and power-clean—have long been used as core exercises by athletes to enhance performance in sport. (11,27) These multi-joint exercises develop the largest and most powerful muscles of the body and have biomechanical and neuromuscular similarities to many athletic movements, such as running and jumping. Recently CKCE have been used and recommended in clinical environments, such as during knee rehabilitation following anterior cruciate ligament (ACL) reconstruction surgery (22,33,38,43,44,50,67,68).

It is difficult to compare tibiofemoral compressive forces during the squat between various published studies since some studies modeled both external forces (e.g., gravity, ground reaction, inertia) and internal forces (e.g., muscle, bone, ligament) (3,13,36,42), while others modeled only external forces (1,20,58). Furthermore, only three of these studies specified the direction of the tibiofemoral shear force (36,41,58), making it difficult to determine which cruciate ligament was loaded. All three of these studies found moderate posterior cruciate ligament (PCL) tensile forces at higher knee angles (0° = full knee extension) and minimum ACL forces at smaller knee angles. Exact knee angles were stated in only one of these studies (58). Only one known study quantified patellofemoral compressive forces during the squat exercise (46). However, the squats in

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this study were performed isometrically. There are no known studies that have quantified tibiofemoral or patellofemoral compressive forces during a dynamic leg press exercise, although Steinkamp et al. (55) did quantify patellofemoral compressive forces during an isometric leg press at 0°, 30°, 60°, and 90° knee flexion.

OKCE, such as seated knee extension and knee flexion exercises, are viewed as single joint, single muscle group exercises. These exercises appear to be less functional in terms of many athletic movements and primarily serve a supportive role in strength and conditioning programs. Moreover, the use of OKCE in clinical settings appears to be diminishing (44,50).

Two known studies have quantified patellofemoral compressive forces during the knee extension exercise. Kaufman et al. (26) quantified patellofemoral compressive force during a dynamic knee extension, while Steinkamp et al. (55) quantified patellofemoral compressive forces during an isometric knee extension at 0°, 30°, 60°, and 90° knee flexion.

Several isometric (7,22,33,41,69) and dynamic (26,67) studies have shown that during the knee extension exercise, the ACL is loaded at knee angles less than 60°, increasing as knee angle decreases. Conversely, the posterior cruciate ligament (PCL) is loaded at knee angles greater than 60°.

Understanding and comparing knee forces and muscle activity in different exercises is essential for determining how to achieve optimal balance of muscle force, ligament tension, and joint compression. Lutz et al. (33) compared knee forces and muscle activity in CKCE (simulated "leg press" in an upright position, as in performing a step-up exercise) and OKCE (knee extension and knee flexion), but these exercises were performed isometrically. In our preliminary study, tibiofemoral compressive forces and muscle activity during dynamic CKCE (leg press, squat) and OKCE (knee extension) were quantified and compared (64). While the study reported tibiofemoral compressive and shear forces, the model did not consider differences between patellar tendon force and quadriceps tendon force; furthermore, tensile forces in the PCL, ACL, and patellofemoral joint were not quantified. Hence, no study has thoroughly described knee biomechanics during dynamic CKCE and OKCE. The purpose of this study was to quantify and compare cruciate ligament tensile forces, tibiofemoral compressive forces, patellofemoral compressive forces, and muscle activity about the knee during dynamic CKCE and OKCE. Internal muscle forces were calculated to estimate the actual forces across the articulating surfaces of the knee.

MATERIALS AND METHODS

Subjects. Ten male subjects experienced in weight training served as subjects. This population was chosen because they specialized in performing the squat, leg press, and knee extension exercises. Since the objectives of this study were to compare knee forces and muscle activity between exercises, it was important to choose experienced subjects who could perform these exercises correctly throughout a full range of knee flexion (i.e., approximately

0–90° knee flexion range). The subjects had a mean height of 177 ± 9 cm, a mean mass of 93 ± 15 kg, and a mean age of 29 ± 6 yr. All subjects performed CKCE and OKCE regularly in training and had no history of knee injuries or knee surgery. Before participating in the study, informed consent was obtained from each subject. Bilateral symmetry was assumed, thus force, video, and electromyographic (EMG) data were collected and analyzed on the subject's left side.

Testing setup. Each subject was tested performing two CKCE (the squat and leg press) and one OKCE (knee extension). A standard 20.5 kg Olympic barbell, disks (Standard Barbell) and a Continental squat rack were used during the squat. Each subject squatted with his left foot on an AMTI (Model OR6–6-2000, Advanced Mechanical Technologies, Inc., Watertown, MA) force platform, and his right foot on a solid block (Fig. 1).

A variable resistance leg press machine (Model MD-117, Body Master, Inc., Rayne, LA) was used during the leg press CKCE. An AMTI force platform for the left foot and a solid block for the right foot were mounted on a customized leg press plate as shown in Figure 2. The force platform, solid block, and leg press plate all remained stationary throughout the lift, while the body moved away from the feet.

A Hoggan variable resistance seated knee extension machine (Model 2055, Hoggan Health Industries, Draper, VT) was used during the knee extension OKCE. A load cell (Model LCCA-500, Omega Engineering, Inc., Stamford, CT) was installed to directly measure force applied by the left leg onto a resistance pad (Fig. 3).

Spherical plastic balls (3.8 cm in diameter) covered with reflective tape were attached to adhesives and positioned over the following bony landmarks: medial and lateral malleoli of the left foot, upper edges of the medial and lateral tibial plateau of the left knee, posterior superior greater trochanters of the left and right femurs, and lateral acromion of the left shoulder. In addition, a 1 cm² piece of reflective tape was positioned on the third metatarsal head of the left foot.

Four electronically synchronized high-speed charged couple device (CCD) cameras (Motion Analysis Corpora-

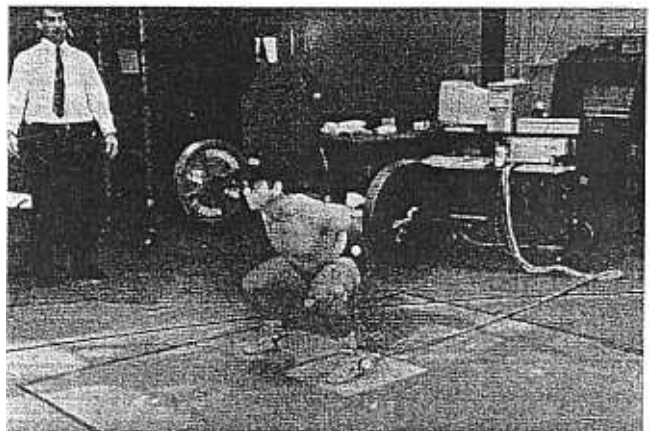


Figure 1—Testing setup for squat exercise.

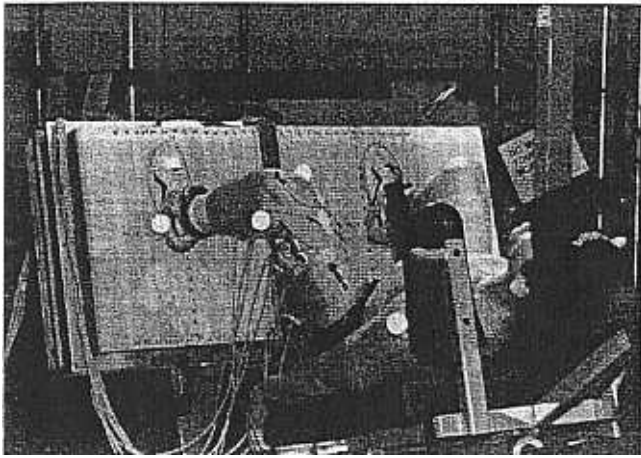


Figure 2—Testing setup for leg press exercise.

tion, Santa Rosa, CA) were strategically positioned around each subject. These cameras collected 60 Hz video data from the reflective markers positioned on the body. Images from these cameras were transmitted directly into a motion analysis system (Motion Analysis Corporation).

EMG data from the quadriceps, hamstrings, and gastrocnemius musculature were quantified with an eight channel, fixed cable, Noraxon Myosystem 2000 EMG U (Noraxon USA, Inc., Scottsdale, AZ). The amplifier bandwidth frequency ranged from 15–500 Hz, (14,65) with an input voltage of 12 VDC at 1.5 A. The input impedance of the amplifier was 20,000 k Ω , and the amplitude of the raw EMG as recorded at the electrodes was expressed in millivolts. The common-mode rejection ratio was 130 Db.

The skin was prepared by shaving, abrading, and cleaning. A model 1089 mk II Checkrode electrode tester (UFI, Morro Bay, CA) was used to test the contact impedance between the electrodes and the skin, with impedance values less than 200 k Ω considered acceptable (14). Most impedance values were less than 10 k Ω .

Blue Sensor (Medicotest Marketing, Inc., Ballwin, MO) disposable surface electrodes (type N-00-S) were used to collect EMG data. These oval shaped electrodes (22 mm wide and 30 mm long) were placed in pairs along the longitudinal axis of each muscle or muscle group tested, with a center-to-center distance between each electrode of approximately 2–3 cm. One electrode pair was placed on each of the following muscle locations in accordance with procedures from Basmajian and Blumenstein (6): 1) rectus femoris, 2) vastus lateralis, 3) vastus medialis, 4) biceps femoris, 5) medial hamstrings (semimembranosus/semitendinosus), and 6) gastrocnemius.

EMG, force, and video data collection equipment were electronically synchronized. EMG and force data were collected by an ADS analog-to-digital system (Motion Analysis Corporation) at 960 Hz. The 960 Hz sampling rate was chosen to time match the EMG and force data with the 60 Hz video data.

Data collection. Each subject came in for a pretest 1 wk before the actual testing session. At this time the experimental protocol was reviewed and the subjects were given

the opportunity to ask questions. In addition, a subject's 12 repetition maximum (12 RM) was determined for each exercise by using the most weight he could lift for 12 consecutive repetitions. The mean 12 RM loads lifted during the squat, leg press, and knee extension were 146.5 ± 39.0 kg, 146.0 ± 30.3 kg, and 78.6 ± 18.2 kg, respectively. While performing the squat and leg press during both the pretest and the actual testing session, each subject used a stance and foot position normally used in training.

Before the testing session began, the force platforms and load cell were calibrated and their positions were determined. To determine three-dimensional locations of the force platforms, video data were collected from 2 cm² pieces of reflective tape positioned on each of the four corners of both force platforms. The three-dimensional locations of each corner of the force platform were then derived in global coordinates. For the knee extension exercise, a reflective marker was permanently attached to the load cell. Therefore, the location of the foot relative to the force platform or load cell and the location of the three-dimensional reaction force vector acting on the foot or leg were able to be determined. All three exercises occupied the same filming area; consequently, video and force data were collected from all trials (i.e., repetitions) from one exercise before setting up for the next exercise. The order of performing the exercises was randomly assigned for each subject. Testing procedures were explained to each subject before testing commenced. Each subject was allowed to perform as many warm-up sets as needed; however, to prevent fatigue, the subjects were instructed not to warm up in excess of 60% of their 12 RM pretest weight. For both the warm-up and testing sets, each subject rested long enough until he felt completely recovered from the previous set. Because of the submaximal weight lifted, the low sets and repetitions performed, and the high fitness level of the subjects, fatigue was assumed to be negligible.

Each subject's stance width in CKCE was measured with a grid overlaid on the squat and leg press force platforms. The mean stance width (inside heel to inside heel) was 40 ± 8 cm for the squat and 34 ± 14 cm for the leg press. A goniometer was used to measure forefoot abduction (i.e.,

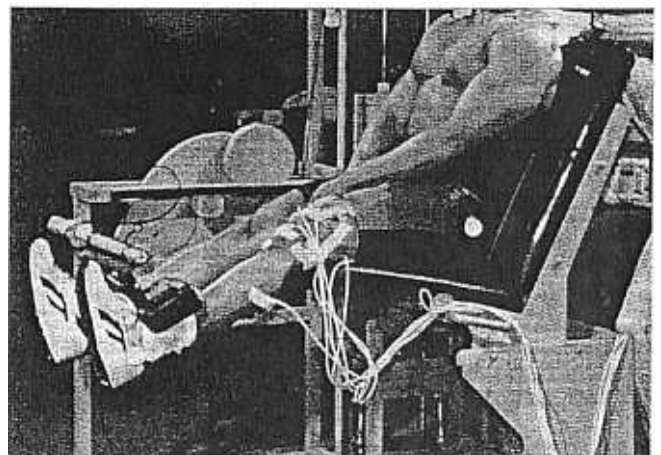


Figure 3—Testing setup for knee extension exercise.

how far the feet were turned outward from the straight ahead position). The mean foot angle was $22^\circ \pm 11^\circ$ for the squat and $18^\circ \pm 12^\circ$ for the leg press. Once the feet were appropriately positioned for the squat and leg press, a tester gave a verbal command to begin the exercise.

Each exercise was performed in a slow and continuous manner. For all subjects, knee flexion and knee extension rates were similar during all exercises, thus minimizing any inertial effects due to cadence. For all subjects and all exercises, the knee flexing phase ranged approximately from 1.5–2 s, while the knee extending phase ranged approximately from 1–1.5 s. Because of the consistent cadence of the subjects for all exercises, a subject's knee flexing and knee extending cadence was not controlled.

The beginning and ending position for the squat and leg press was with the knee near full extension. Knee angle was defined as 0° in this fully extended knee position. In a continuous motion the subject descended to maximum knee flexion (approximately 90° – 100°) and then ascended back to the starting position. The starting and ending positions for the OKCE were seated with approximately 90° – 100° knee angle. From the starting position, each subject extended the knees and then returned back to the starting position. The inside heel to inside heel distance in OKCE was approximately 20 cm for all subjects.

Each subject performed one set of four repetitions for each exercise. The first repetition of each set was used to allow the subjects to establish a "groove"; thus data were not collected. Data collection was initiated at the end of the first repetition and continued throughout the final three repetitions of each set. Between each repetition, the subjects were instructed to pause approximately 1 s to provide a clear separation between repetitions.

Subsequent to completing all exercise trials, EMG data were collected during maximum voluntary isometric contractions (MVIC) to normalize the EMG data collected in CKCE and OKCE. Pilot work was conducted before testing to determine the knee and hip positions that produced the greatest possible muscle activity. The MVIC for the rectus femoris, vastus lateralis, and vastus medialis were collected at a position of 90° knee and hip flexion (i.e., 90° – 90° position) while performing the seated knee extension exercise. The MVIC for the lateral and medial hamstrings were collected while performing a seated knee flexion exercise in the 90° – 90° position. MVIC for the gastrocnemius was determined using the leg press while at a position of 0° knee and hip flexion with the feet halfway between the neutral position and maximum plantar flexion. Three 3-s trials were collected for each MVIC, which were also performed in a randomized manner.

Data reduction. Video images for each reflective marker were automatically digitized in three-dimensional space with Motion Analysis ExpertVision software, utilizing the direct linear transformation method (62). Testing of the accuracy of the calibration system resulted in reflective balls that could be located in three-dimensional space with an error less than 1.0 cm. The raw position data were smoothed with a double-pass fourth order Butterworth low-

pass filter with a cut-off frequency of 6 Hz. (49) Using principles of vector algebra and finite difference methods (37), a computer program calculated joint angles, linear and angular velocities, and linear and angular accelerations.

EMG data for each MVIC trial and each test trial were rectified and averaged in a 0.01-s moving window (i.e., linear envelope). Data for each test trial were then expressed as a percentage of the maximum value in the subject's corresponding MVIC trial. EMG, force platform, and load transducer data were reduced from 960 Hz to 60 Hz by retaining only those points which corresponded in time with the video data collected (i.e., every 16th data point).

Calculation of resultant force and torque. The ankle joint center was defined as the midpoint of the medial and lateral ankle markers, while the foot was defined by a line segment from the ankle joint center to the toe marker. The knee joint center was defined as the midpoint of the medial and lateral knee markers. The hip joint center was defined to be located inward 20% of the distance on the line segment from the left to the right hip marker (9). Mass, center of mass, and moments of inertia for the foot and leg were estimated using previously published data (15,59,65).

Resultant joint forces and torques acting on the foot and leg were calculated using three-dimensional rigid link models of the foot and leg and principles of inverse dynamics. Free body diagrams of the foot and leg including all external forces and torques acting on each segment are shown in Figure 4. Inertial force was the product of mass and linear acceleration, while inertial torque was the product of moment of inertia and angular acceleration. External forces were measured directly with the force platforms and load cell. Resultant force applied by the thigh to the leg was separated into three orthogonal components; however, because of the small magnitudes of mediolateral forces observed, only axial compressive and anteroposterior shear forces were analyzed. An anterior shear force was defined as an anterior force the thigh applied to the leg to resist pos-

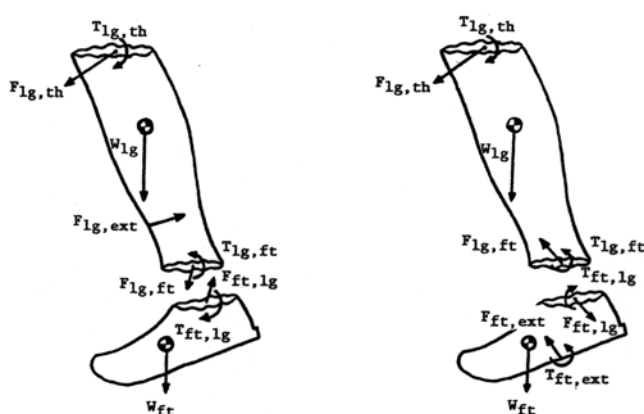


Figure 4—Free-body diagram for (a) open kinetic chain exercise and (b) closed kinetic chain exercise: (W_n) force applied by gravity onto foot; (W_{lg}) force applied by gravity onto leg; (F_{ext}) force applied by force plate or load cell; (T_{ext}) torque applied by force plate onto foot; ($F_{n,lg}$) force applied by foot onto leg; ($F_{lg,n}$) force applied by leg onto foot; ($T_{n,lg}$) torque applied by foot onto leg; ($T_{lg,n}$) torque applied by leg onto foot; (F_{res}) force applied by thigh onto leg; and (T_{res}) torque applied by thigh onto leg.

terior translation of the leg, while a posterior shear force was a posterior force the thigh applied to the leg to resist anterior translation of the leg (33). An anterior shear force is resisted primarily by the PCL, while a posterior shear force is resisted primarily by the ACL (10). Unfortunately, anterior and posterior shear force definitions are inconsistent among studies (26,33,55). Resultant torque applied by the thigh to the leg was separated into three orthogonal components. Because of the small magnitudes in valgus-varus torque and internal-external rotation torque, only extension-flexion torque was analyzed. Resultant force, torque, and EMG data were then expressed as functions of knee angle. For each trial, data from the three repetitions were averaged.

Model for ligament and bone force. To estimate tibiofemoral compressive forces, cruciate tensile forces, and patellofemoral compressive forces in OKCE and CKCE, a biomechanical model of the sagittal plane of the knee was developed (Fig. 5). Since the lateral and medial collateral ligaments play minor roles in stabilizing the knee joint during knee flexion and extension, they were not included in this model.

Because of the slow speed of muscle contraction during the exercises performed, the total force (F) produced by a muscle was assumed to be equal to the product of the maximum force the muscle could produce and EMG activity expressed as a fraction of the maximum EMG value (MEMG) recorded during MVIC. Maximum muscle force was equal to the product of physiological cross-sectional area (PCSA) and maximum voluntary contraction force per PCSA (σ). Hence, $F = (\sigma * PCSA) * (EMG/MEMG)$.

Maximum voluntary contraction force per PCSA was assumed to be $40 \text{ N} \cdot \text{cm}^{-2}$ for the quadriceps and $35 \text{ N} \cdot \text{cm}^{-2}$ for the hamstring and gastrocnemius (12,21,25,39,40). PCSA data from Wickiewicz et al. (63) were used to determine the ratios of PCSA between different muscles. Using these ratios and the 160 cm^2 quadriceps area reported by

Narici et al. (40), PCSA for each muscle was calculated. These PCSA were then scaled for each individual subject by using the ratio of the subject's body weight and the average 75 kg body weight reported by Narici et al. (40)

Tensile force in the quadriceps tendon was the summation of all four quadriceps forces. To calculate force generated in the vastus intermedius, the average of EMG data from the other three quadriceps was used. Since the patellar tendon force changes with knee flexion and extension, tensile force in the patellar tendon was calculated as a function of patellar tendon force and knee angle (60,61). Torque created by each muscle or tendon was the product of the its moment arm (23) and its force. Assuming that ligaments and bones created negligible torque at the knee, the resultant torque should equal the summation of torque produced by the patellar tendon, medial hamstrings, biceps femoris, and gastrocnemius:

$$\vec{T}_{\text{res}} = \vec{T}_{\text{pt}} + \vec{T}_{\text{mh}} + \vec{T}_{\text{bf}} + \vec{T}_{\text{g}}$$

Since the accuracy of estimating muscle forces depends on accurate estimation of PCSA, maximum voluntary contraction force per unit PCSA, and the EMG-force relationship, the torque equilibrium equation shown above may not be satisfied. Therefore, the total force (F) was modified by a coefficient (c): $F = c * (\sigma * PCSA) * (EMG/MEMG)$.

Values for each muscle's coefficient were determined with the optimization routine presented below. Each coefficient was initially set at one and adjusted with the Davidon-Fletcher-Powell algorithm. (45) With this algorithm, coefficients were constrained by an upper and lower limit and were determined so that the summation of muscle torque ($\sum T_m$) equaled the resultant torque.

$$\min f(c_i) = \sum_{i=1}^{nm} \quad + \lambda \left(T_{\text{res}} - \sum_{i=1}^{nm} T_{m_i} \right)$$

Once muscle forces were corrected, tibiofemoral compressive force and PCL/ACL tensile force were found using the following force equilibrium equations:

$$\vec{F}_{\text{res}} = \vec{F}_{\text{tr}} + \vec{F}_{\text{pcl}} + \vec{F}_{\text{acl}} + \vec{F}_{\text{pt}} + \vec{F}_{\text{mh}} + \vec{F}_{\text{bf}} + \vec{F}_{\text{g}}$$

or.

$$\vec{F}_{\text{tr}} + \vec{F}_{\text{pcl}} + \vec{F}_{\text{acl}} = \vec{F}_{\text{res}} - \vec{F}_{\text{mh}} - \vec{F}_{\text{bf}} - \vec{F}_{\text{g}}$$

Tibiofemoral compressive force was assumed to be in the longitudinal direction of the tibia. Cruciate ligament orientation was determined as a function of knee angle using regression equations (23). Tibiofemoral compressive force was constrained to be compression and ligament forces were constrained to be in tension.

Based upon the free-body diagram for the patella (Fig. 5b), patellofemoral compressive force was a function of patellar tendon force and quadriceps tendon force. The angles between the patellar tendon, quadriceps tendon, and patellofemoral joint were expressed as functions of knee angle (60,61).

Statistical analysis. To determine significant differences among the exercise types (knee extension, leg press

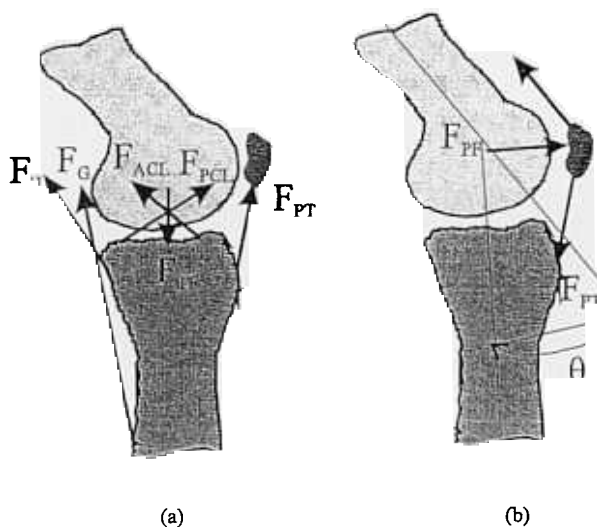


Figure 5—Forces acting on the (a) proximal tibia and (b) patella: (F_h) hamstring, (F_g) gastrocnemius, (F_{acl}) ACL, (F_{pcl}) PCL, (F_{tr}) tibiofemoral, (F_{pt}) patellar tendon, (F_{pfl}) patellofemoral, and (F_{qt}) quadriceps tendon. Knee angle (θ) also shown.

and squats) and phase (knee flexing, knee extending), muscle activity, PCL/ACL tensile force, tibiofemoral compressive force, and patellofemoral compressive force were analyzed every 2° of knee angle with a two factor repeated measure ANOVA ($P < 0.05$). Because of the large number of comparisons and the increased probability of Type I errors, consistency of significant differences as a function of knee angle was paramount. Hence, only significant differences that occurred over three consecutive 2° knee angle intervals were reported in the results. The Student-Newman-Keuls tests were conducted to isolate differences among different comparisons. The tests were repeated for each knee angle analyzed. For graphical presentation, data for all subjects performing each type of exercise were averaged.

RESULTS

Muscle activity. All three quadriceps muscles tested demonstrated similar patterns (Fig. 6). Quadriceps activity was significantly greater in OKCE between 15–65° knee angle, while quadriceps activity was significantly greater in CKCE at knee angles greater than 83° (Table 1). Hamstring activity remained low throughout the leg press and knee extensions (Fig. 7) and showed no significant differences (Table 1). Throughout, knee extending the squat generated significantly greater lateral hamstring activity than the leg press and knee extensions, while no significant differences were observed during knee flexing (Table 1). No significant differences were observed in the medial hamstrings for all exercises. Gastrocnemius activity was similar to quadriceps

TABLE 1. Significant differences in muscle activity among the knee extension (KE), leg press (LP), and squat (SQ) exercises.

Muscle	Phase	Knee Angle (deg)	Significant Difference ($P < 0.05$)
Rectus Femoris	Knee flexing	15–65	KE > SQ & LP
		83–95	SQ & LP > KE
	Knee extending	83–95 15–57	SQ & LP > KE KE > SQ & LP
Vastus medialis	Knee flexing	15–45	KE > SQ & LP
		71–95	SQ & LP > KE
	Knee extending	59–95 15–33	SQ & LP > KE KE > SQ & LP
Vastus lateralis	Knee flexing	15–45	KE > SQ & LP
		69–75	SQ > KE
		75–95	SQ & LP > KE
	Knee extending	70–95	SQ & LP > KE
		55–70	SQ > KE
		29–37 15–29	KE > LP KE > SQ & LP
Biceps femoris	Knee flexing	No significant differences	
	Knee extending	90–95	SQ > KE
		27–90	SQ > LP & KE
Medial hamstrings	Knee flexing	No significant differences	
	Knee extending	No significant differences	
Gastrocnemius	Knee flexing	15–29	KE > SQ & LP
		73–95	SQ & LP > KE
	Knee extending	69–95	SQ & LP > KE
		15–39	KE > SQ & LP

activity (Fig. 7). When the knee was near full extension, gastrocnemius activity was significantly greater in OKCE and when the knee was near full flexion, gastrocnemius activity was significantly greater in CKCE.

Resultant forces and torques. Resultant forces and torques reflect external and inertial forces only, with internal muscle forces not considered. These data are shown in Figure 8. Approximately 1000 N of tibiofemoral compressive force was produced throughout the CKCE. Minimal levels of distractive force (negative compressive force) were produced throughout OKCE. Anterior shear force in CKCE increased with knee angle, peaking at approximately 600 N during knee extending. In OKCE, anterior shear force was greatest in the mid-range of knee angle, peaking at approximately 400 N during knee extending.

The greatest extension torque about the knee was produced during the mid-range of knee extending in OKCE, peaking at approximately 200 N·m. Peak torque in CKCE was approximately 175 N·m, and occurred near full knee flexion during knee extending. Extensor knee torque values progressively increased throughout knee flexing and progressively decreased throughout knee extending.

Tibiofemoral compressive forces. With internal muscle forces considered, tibiofemoral compressive forces were approximately three times greater than resultant compressive forces (Figs. 8 and 9). Between 15–29° knee angle, tibiofemoral compressive forces were greatest in OKCE during both knee flexing and knee extending (Fig. 9 and Table 2). Between 71–95° knee angle during knee flexing, tibiofemoral compressive forces were greatest in CKCE. For

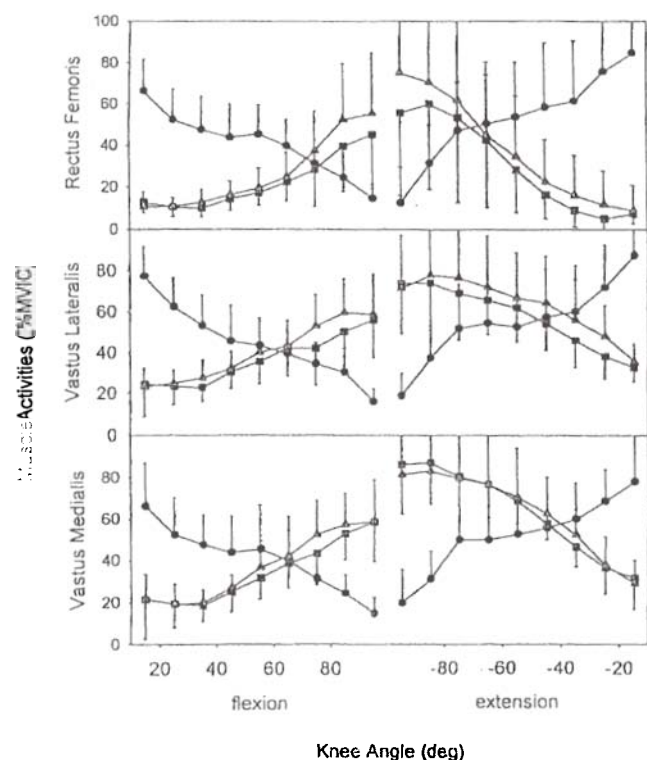


Figure 6—Mean and SD of quadriceps muscle activity during squat (▲), leg press (■), and knee extension (●), expressed in percentage of maximum voluntary isometric contraction (%MVIC).

all exercises, approximately 3000 N of maximum tibiofemoral compressive force was produced (Table 3). Maximum tibiofemoral compressive force was produced between 53–93° knee angle in CKCE and between 39–57° in OKCE (Table 3).

PCL/ACL tensile forces. For all exercises, PCL tensile forces generally increased with knee flexion and decreased with knee extension (Fig. 9). In CKCE, the PCL was always in tension. In OKCE, the PCL was in tension when the knee angle was greater than 25°, while the ACL was in tension when the knee was near full extension (15–25°). Peak PCL tensile forces were approximately 2000 N in CKCE and approximately 1000 N in OKCE (Table 3).

Patellofemoral compressive forces. Patellofemoral compressive forces generally increased with knee flexion and decreased with knee extension (Fig. 9). However, in OKCE patellofemoral compressive force decreased near full flexion. OKCE produced significantly greater forces than CKCE at knee angles less than 57°, while CKCE generated significantly larger forces than OKCE at knee angles greater than 85° (Table 2). Maximum patellofemoral compressive force was between 4000–5000 N for all three exercises (Table 3).

DISCUSSION

The aim of this study was to compare knee biomechanics during dynamic OKCE and CKCE throughout a continuous range of motion. Both the knee flexing and knee extending portions of each exercise were examined. Muscle activity for all of the major knee muscles were measured. Resultant

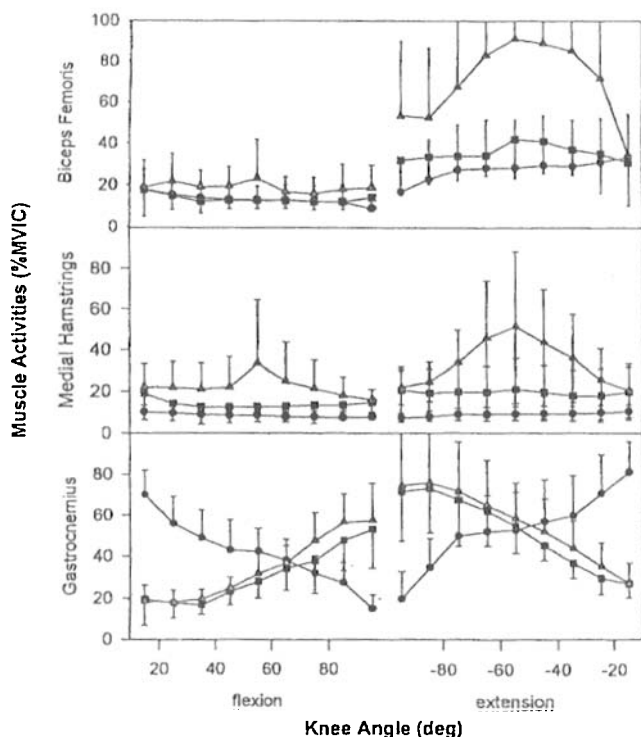


Figure 7—Mean and SD of hamstring and gastrocnemius muscle activity during squat (▲), leg press (■), and knee extension (●), expressed in percentage of maximum voluntary isometric contraction (%MVIC).

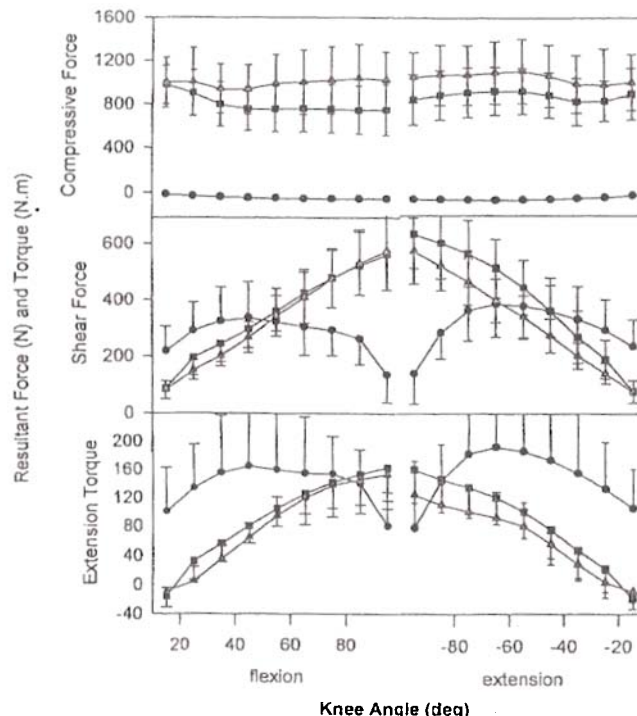


Figure 8—Mean and SD of resultant force and torque during squat (▲), leg press (■), and knee extension (●). Compressive force, anterior shear force, and extension(+)/flexion(−) torque are shown.

joint forces and torques were calculated, but these calculations considered only the external and inertial forces and torques acting on the foot and leg. To identify the contribution of individual ligaments and articulations, a biomechanical model of the knee was developed modeling internal muscle forces and torques. While this model has numerous uncertainties associated with current biomechanical techniques, the results provide valuable insight regarding specific hard and soft tissue structures.

It is difficult to compare results with other studies because of methodological variances among studies. Several studies involved maximum isometric contractions at select angles, (33,42,43,46,53,55,69), while other studies involved dynamic movements (3,13,26,36,41,58). Furthermore, none of these dynamic studies specified the percent of each subject's maximum load in which they performed these exercises. In this study a typical 12 RM intensity was employed, which is approximately equivalent to 70–75% of each subject's 1 RM (35). Performing 8–12 repetitions is a common repetition scheme that many physical therapy, athletic training, and athletic programs adhere to for strength development (56,57). Since the same relative weight was used for all exercises (i.e., 12 RM), ligamentous tensile forces and tibiofemoral and patellofemoral compressive forces were able to be compared with each other.

From Table 3, the SD among maximum tibiofemoral compressive forces, ACL and PCL tensile forces, and patellofemoral compressive forces were quite high. This was largely a result of the high variability in each subject's 12 RM. In these well trained lifters, those subjects with higher body weight usually had a higher 12 RM than subjects with

lower body weight. The subjects' body weight ranged from approximately 70–110 kg, while their 12 RM squat ranged from approximately 100–220 kg, their 12 RM leg press ranged from approximately 100–180 kg, and their 12 RM leg extension ranged from approximately 60–90 kg.

Muscle activity. Averaging over the entire exercise, OKCE generated approximately 45% more rectus femoris activity than CKCE, while CKCE generated approximately 20% more vastus medialis activity and approximately 5% more vastus lateralis activity than OKCE. These findings are in agreement with Signorile et al. (52) who found significantly more vasti activity during the squat exercise than during the knee extension exercise. This suggests that OKCE may be more effective in developing the rectus femoris, while CKCE may be more effective in developing the vasti muscles. However, this may be true only at specific ranges of knee motion. From Table 1, rectus femoris activity was significantly greater in OKCE at knee angles less than 65°, while CKCE produced more rectus femoris activity between 83–95° knee angle. Similarly, vasti activity was greater in OKCE at knee angles less than 45°, while CKCE produced more activity at knee angles greater than 55°. Comparing muscle activity in OKCE, the vastus medialis, vastus lateralis, and rectus femoris all generated a similar amount of muscle activity. In a comparison of muscle activity in CKCE, the two vasti muscles produced approximately 50% greater activity than the rectus femoris, which is in accordance with squat data from Wretenberg et al. (66) Furthermore, the vastus medialis and lateralis generated approximately the same amount of muscle activity, which is in agreement with squat data from Signorile et al. (52). These findings have important clinical implications when

TABLE 2. Significant differences in PCL tensile forces, tibiofemoral compressive forces, and patellofemoral compressive forces among the knee extension (KE), leg press (LP), and squat (SQ) exercises.

Force	Phase	Knee Angle (deg)	Significant Difference ($P < 0.05$)
PCL	Knee flexing	15–29	KE > SQ & LP
		71–95	SQ & LP > KE
	Knee extending	15–25	KE > SQ
		15–27	KE > LP
	Knee flexing	15–33	LP > KE
		33–45	SQ & LP > KE
		61–69	SQ & LP > KE
		69–95	LP > KE
Patellofemoral	Knee extending	15–95	SQ & LP > KE
		27–79	SQ > LP
	Knee flexing	15–47	KE > SQ & LP
		85–95	SQ & LP > KE
	Knee extending	15–57	KE > SQ & LP
		89–95	SQ & LP > KE

one is deciding which exercise modality to choose during knee rehabilitation. For overall quadriceps development, OKCE may be superior or at least as effective as CKCE. However, a major concern for therapists during knee rehabilitation is muscle imbalances between the vasti muscles. These imbalances can cause patellar tracking dysfunction, which can result in patellar subluxation, patellar tendinitis, or chondromalacia patellar. It has been shown that the vastus medialis is the first muscle of the quadriceps group to atrophy after injury or non-use, and it responds to therapy slower than the vastus lateralis (18,19,32,47). Since overall vastus medialis activity was greater in CKCE, these closed chain exercises may be superior to or at least as effective as OKCE in maintaining muscle balance between the vasti muscles. In a comparison of overall quadriceps activity between the squat and leg press, the squat was slightly more effective in generating rectus femoris, vastus medialis, and vastus lateralis activity.

Numerous studies have shown that the EMG magnitude with eccentric work is much less than the EMG magnitude during an equal amount of concentric work (4,8,28,29,58). This was true in this study, as quadriceps activity was lower during knee flexing (eccentric work) than during knee extending (concentric work).

Previous studies have demonstrated that co-contraction between the quadriceps and hamstrings occur in OKCE (5,17). These studies hypothesized that co-contraction between the quadriceps and hamstrings help stabilize the knee and thereby minimize potential tensile loading to the ACL. Similar to data from Lutz et al. (33), this study found greater co-contraction between the quadriceps and hamstrings in CKCE compared with that in OKCE. The greatest difference in hamstring activity between CKCE and OKCE occurred during knee extending.

Figure 7 shows that peak hamstring activity during the squat was approximately 35% of a MVIC during the knee flexing phase and approximately 50% of a MVIC during the knee extending phase, with peak values occurring near 50° knee angle during both phases. In contrast, peak hamstring

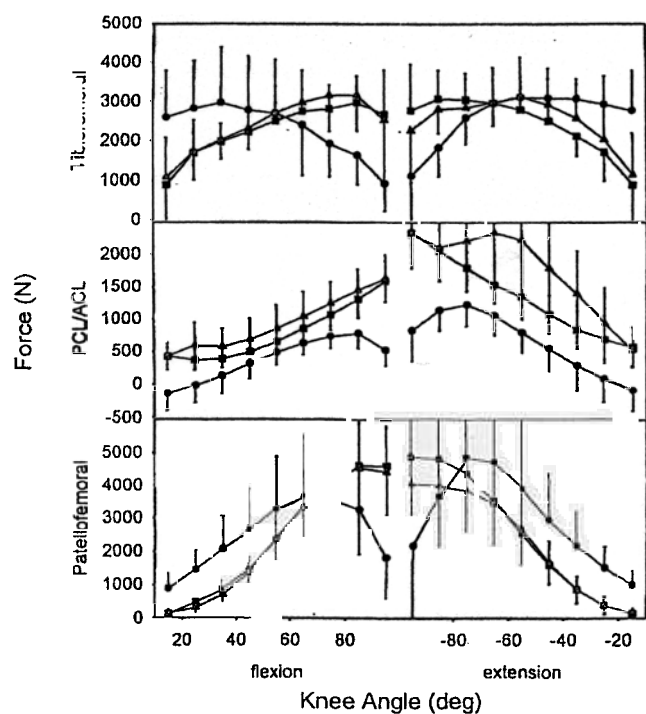


Figure 9—Mean and SD of forces during squat (▲), leg press (■), and knee extension (●). Tibiofemoral compressive force, PCL(+)/ACL(-) tensile force, and patellofemoral compressive force are shown.

TABLE 3. Maximum ACL/PCL tensile forces, tibiofemoral compressive forces, and patellofemoral compressive forces (N) during knee flexing and knee extending. For each parameter, the Mean $\pm$ SD for the group of subjects is shown, as well as the corresponding mean knee angle.

Force	Phase	Squat	Leg Press	Knee Extension
Tibiofemoral	Knee flexing	2192 $\pm$ 930 @ 81°	3011 $\pm$ 693 @ 93°	3017 $\pm$ 1511 @ 39°
	Knee extending	3134 $\pm$ 1040 @ 53°	3155 $\pm$ 755 @ 91°	3285 $\pm$ 1927 @ 57°
PCL	Knee flexing	1635 $\pm$ 369 @ 95°	1593 $\pm$ 316 @ 95°	801 $\pm$ 221 @ 83°
	Knee extending	1868 $\pm$ 878 @ 63°	1866 $\pm$ 383 @ 95°	959 $\pm$ 300 @ 79°
ACL	Knee flexing	0	0	142 $\pm$ 258 @ 15°
	Knee extending	0	0	158 $\pm$ 256 @ 15°
Patellofemoral	Knee flexing	4548 $\pm$ 1395 @ 85°	4780 $\pm$ 1194 @ 91°	3724 $\pm$ 1940 @ 87°
	Knee extending	4042 $\pm$ 955 @ 95°	4991 $\pm$ 1352 @ 91°	4846 $\pm$ 2453 @ 75°

activity from squat data from Stuart et al. (58) was approximately 20% of a MVIC during both the knee flexing and knee extending phases, with peak values occurring near 30° knee angle during both phases. These lower EMG hamstring magnitudes by Stuart et al. are probably a result of their subjects lifting a lower percentage of their 1 RM compared with subjects in the current study. The similar hamstring activity they observed between the knee flexing and knee extending phases of the lift is contrary to the results from the current study, which showed significantly greater hamstring activity during knee extending. Since the hamstrings are biarticulate muscles, it is difficult to delineate these muscles during the squat as performing eccentric work during knee flexing and concentric work during knee extending. They may actually be working isometrically during both phases of the squat, since they are shortening at the knee and lengthening at the hip during knee flexing and lengthening at the knee and shortening at the hip during knee extending. If they are indeed working eccentrically during knee flexing and concentrically during knee extending, as is traditionally believed, then our results would be in accord with other studies that have shown decreased activity during eccentric work and increased activity during concentric work (8,29).

Data averaged during the entire phase shows that the leg press produced slightly more hamstring activity than OKCE, while the squat produced approximately twice as much hamstring muscle activity as the leg press and OKCE. Consequently, the squats may be more effective in hamstring development than the leg press and leg extensions. The greater hamstring activity produced during the squat exercise was primarily a result of the hamstrings role in controlling hip flexion during knee flexing and producing hip extension during knee extending.

During the leg press and OKCE, a relatively small flexor torque is generated about the hip; therefore, minimal hamstring activity is need to extend the hip (44). The antagonistic hamstring activity during the squat provides greater stability against anterior displacement of the leg relative to the thigh, thus reducing potential tension in the ACL and increasing tension in the PCL. This is consistent with the findings of the current study. During the mid-range of knee extending in the squat when hamstring activity was greatest, PCL tension was also greatest. A similar pattern of higher hamstring activity and greater anterior shear force (i.e., PCL tensile force) during the knee extending phase of the dy-

namic barbell squat has also been observed by Stuart et al. (58).

In CKCE the gastrocnemius contracted eccentrically to control the rate of dorsiflexion during knee flexing and contracted concentrically to cause plantar flexion during knee extending. Since the foot was free to move and was not restrained in OKCE, minimal gastrocnemius activity was presumed. On the contrary, higher than expected values were observed throughout the range of knee motion. This higher activity may be caused by a propensity to plantar flex the ankle while performing the knee extension exercise. A more plausible explanation is that the biarticulate gastrocnemius co-contracted with the hamstrings to help stabilize the knee while performing the OKCE. Since the hamstrings and gastrocnemius both cross the knee posteriorly, they provide posterior knee stabilization during knee movements. Since a shear force component from the patellar tendon attempts to translate the leg anteriorly relative to the thigh at knee angles less than 60°, (26,67) the higher gastrocnemius activity observed at lower knee angles may help resist this translation.

Resultant forces and torques. Resultant compressive forces were equal to 1.1 times body weight (BW) in CKCE and nonexistent in OKCE. It is still unclear when compressive force magnitudes become detrimental to the knee joint. The maximum compressive force of 1.1 times BW in CKCE is considerably less than the maximum compressive force of 2.0 times BW that has been calculated during slow running at 3 m·s⁻¹ (2).

Resultant shear force direction is important since it provides insight concerning tensile loading to the cruciate ligaments. Butler et al. (10) have shown that the ACL provides 86% of the total resistance to anterior drawer and the posterior cruciate ligament (PCL) provides approximately 95% of the total restraining force to posterior drawer. Two squat studies found shear force magnitudes that were similar to those found in the current study (1,36). Of these, only Meglan et al. (36) specified shear force direction. Like the results from this study, they found anterior shear forces (i.e., PCL tensile force) throughout the knee flexing and knee extending phases of the squat. Stuart et al. (58) also observed PCL tensile forces caused by shear forces generated during the dynamic barbell squat. Similar to the current study, the shear forces generated during the squat progressively increased throughout knee flexing and progressively

decreased throughout knee extending. The higher resultant shear force magnitudes from the current study compared with the magnitudes in Stuart et al. is primarily because the subjects from the current study used a higher percent of their 1 RM. Some physicians, therapists, and coaches feel that large shear forces produced in CKCE and OKCE may have deleterious effects on the knee. However, maximum anterior shear forces were only 0.67 times BW in CKCE and 0.44 times BW in OKCE. This is considerably less than the maximum anterior shear force of 1.0 times BW that has been reported during slow running at 3 m·s⁻¹ (2). Furthermore, running is often performed at a greater frequency and duration compared with that at CKCE and OKCE, greatly increasing knee injury potential caused by excessive shear forces being applied to the knee during each stride.

Knee extensor torques are generated in CKCE and OKCE primarily to overcome the load being lifted. The quadriceps are the primary muscle group that generates this knee extensor torque, contracting eccentrically during the knee flexing phase to control the rate of knee flexion and concentrically during the knee extending phase to overcome forces due to gravity. Extensor torques values and patterns were similar to values and patterns reported in numerous other studies (30,31,42,58,66). No known studies have reported knee extensor torques during an isotonic leg press or isotonic knee extension exercise.

Tibiofemoral compressive forces. Tibiofemoral compressive forces have been determined to be an important factor in stabilizing the knee by resisting anteroposterior translational movement due to shear forces (24,34,51,68). With internal muscle forces estimated, these forces were approximately three times the resultant tibiofemoral compressive forces (i.e., tibiofemoral compressive forces due to external and inertial forces only). With muscle weakness of fatigue, compressive forces decrease, which may compromise knee stability. Compressive forces may be especially important when the knee is near full flexion, for this is when the greatest PCL tensile forces occurred. It remains unclear how much compressive force is desirable and when it produces adverse effects. When the knee was near full flexion, tibiofemoral compressive forces were greater in CKCE. These data are consistent with results from Lutz et al. (33), which also demonstrated greater compressive forces in CKCE compared with those in OKCE. Furthermore, a similar tibiofemoral compressive force pattern during the barbell squat has been observed by Stuart et al. (58).

PCL/ACL tensile forces. PCL tensile forces were generated in CKCE throughout the knee flexing and knee extending phases and were also generated in OKCE between 25–95° knee angle. Peak force was 1.5 to 2.0 times BW in CKCE and approximately 1.0 times BW in OKCE. These magnitudes and knee angles were similar to shear force results reported in previous studies involving dynamic movement (3,26,36,41), but higher than results in studies involving isometric contractions (33,42,43,53,69). It is difficult to compare PCL tensile forces among studies, since most other studies did not model muscle and cruciate ligamentous forces; hence, the actual articulating forces across

the knee joint cannot be determined. When an individual's PCL is weak, caution should be taken when performing OKCE and CKCE at higher knee angles, since PCL tensile forces were greatest at these positions. PCL tensile forces were greatest for all exercises during knee extending.

Peak ACL tensile forces in OKCE were approximately 0.20 times BW and occurred at 15° knee angle. This magnitude and knee angle were similar to results reported during other studies involving the knee extension exercise (26,33,42,53,69). The large compressive forces produced during these small knee angles may aid the ACL in knee stabilization. The presence of ACL tension because of posterior shear force appears somewhat contradictory, since a resultant anterior shear force (i.e., PCL tensile force) was produced in OKCE. However, muscle force contributions are not included in the resultant force calculations. These forces reflect only the effects of gravity, inertia, and the posteriorly directed external force acting on the leg by the resistance pad. The external force of the resistance pad attempts to translate the leg posteriorly relative to the thigh, which alone would load the PCL. PCL tensile and muscle forces are primarily responsible for resisting this external force by applying an anteriorly directed force to the leg relative to the thigh. The quadriceps, via the patella tendon, exerts an anteriorly directed force on the leg between approximately 0–65° knee angle and a posteriorly directed force when the knee is flexed greater than approximately 60° (23). In contrast, the hamstrings exert a posteriorly directed force throughout the knee range of motion. When the anterior force component of the patella tendon force exceeds the posterior force components of the hamstrings and external resistance, a net anteriorly directed force is applied to the leg, which is primarily resisted by the ACL (10). Since there is much more quadriceps activity than hamstrings activity during the knee extension exercise, the ACL can potentially be loaded at knee angle less than approximately 60°. This ACL loading between 0–60° knee angle has been confirmed experimentally (26,33,42,53,69). For an individual with a weak ACL, caution should be taken when performing OKCE when the knee is near full extension, as this is when ACL loading occurs. This is consistent with previous studies comparing CKCE and OKCE (33,44,48).

Patellofemoral compressive forces. High patellofemoral compressive forces, which can potentially cause high stresses on the undersurface of the articular cartilage of the patella, are believed to be the initiating factors for patellofemoral dysfunction (e.g., chondromalacia) and subsequent osteoarthritis. Magnitudes and knee angles associated with peak force were similar to results reported during other studies involving OKCE and CKCE exercises (13,26,41,46,53,55).

Similar to the current study, Steinkamp et al. (55) had male subjects perform knee extension (OKCE) and leg press (CKCE) exercises using their 10 RM. However, they performed these exercises isometrically at 0°, 30°, 60°, and 90° knee angles. Results between studies produced both similarities and differences. Force patterns between studies were

similar during the leg press, with forces progressively increasing as knee angle increased (Fig. 9). In addition, peak forces during knee extensions were similar and occurred at similar knee angles. Although peak forces also occurred at similar knee angles during the leg press, the peak force from Steinkamp et al. (55) was approximately twice the peak force calculated in the current study. This large discrepancy is surprising, especially since their subjects lifted less weight than the lifters in the current study. The different types of knee extension and leg press machines used among studies may explain some of this variance. How these exercises were performed (i.e., isometric vs dynamic) may also explain some of the incongruity in forces generated. For example, there are no inertial forces during isometric exercise, while inertial forces can exist during dynamic exercise, although they are small when weight training exercises such as the squat are being performed (30). In addition, inertial forces may have affected the shape of the curves from both studies during the knee extension exercise. Although forces during knee extensions increased at lower to mid-range knee angles and decreased at higher knee angles, the slope of these curves are quite different. Force data from Steinkamp et al. is nearly identical at 0°, 30°, and 60° knee angle (approximately 4000 N), increasing only slightly from 0° to 60°, and then dropping sharply to 0 N at 90°. In sharp contrast, force data from the current study was approximately 1000 N, 2000 N, and 4000 N at 15°, 30°, and 60°, respectively. These incongruities can partially be explained by considering the inertial characteristics that exist during the knee extending phase of the knee extension exercise (Fig. 9). Forces were initially low at high knee angles (i.e., at the start of the exercise) as the subjects began exerting force against the resistance pad. Subsequently, from approximately maximum knee flexion to knee mid-range the subjects accelerated the leg and forces increased proportionately. From approximately mid-range until full knee extension, the leg began to accelerate in the opposite direction (i.e., slow down or decelerate) to prevent the knee from forcefully hyper extending; hence, forces decreased proportionately.

In contrast to data from Steinkamp et al. (55), patellofemoral compressive force data from Kaufman et al. (26) during an isokinetic knee extension are remarkably similar (both in shape and magnitudes) to the knee extension patellofemoral compressive force data displayed in Figure 9. Patellofemoral compressive force data from both Kaufman et al. and the current study progressively increased until approximately 70° knee angle, and then progressively decreased as the knee continued flexing. In addition, the 60°/s used by Kaufman et al. was approximately the same rate of knee rotation used by the subjects in the current study. It can be concluded that these two dynamic studies involving the knee extension exercise produced quite a different patellofemoral compressive force pattern compared to knee extension studies involving isometric contractions (46,53,55).

Although patellofemoral compressive force was greatest at higher knee angles during both the knee extensions and leg press, patellofemoral stress (i.e., patellofemoral com-

pressive force per contact area between the patella and femur) may be the most important factor in patellofemoral dysfunctions, such as patellofemoral chondromalacia. Using patellofemoral contact areas of 1.5 cm² for 0° knee angle, 3.1 cm² for 30° knee angle, 3.9 cm² for 60° knee angle, and 4.1 cm² for 90° knee angle, Steinkamp et al. (55) demonstrated that patellofemoral stress was greatest at the lowest knee angle (0°) during the knee extension exercise and greatest at the highest knee angle (90°) during the leg press. However, these data should be interpreted with caution since the patella is not in contact with the femoral trochlea at 0° knee angle (i.e., terminal knee extension). Patellofemoral stress typically begins at approximately 10° knee angle, which is when the patella begins to glide onto the articular surface of the femoral trochlea. Steinkamp et al. (55) further demonstrated that patellofemoral stress was less during the leg press at knee angles less than 48°, which is a more functional knee angle range in human movement and locomotion compared to knee angles between 48° and full knee flexion. Applying these patellofemoral contact areas to data from the current study yielded patellofemoral stress values at comparable knee angles with patellofemoral stress during the leg press progressively increasing as knee angle increased, peaking at approximately 90° knee angle. This is in agreement with data from Steinkamp et al., which displayed the same general pattern of progressive increasing patellofemoral stress as knee angle increased. However, a disparity occurred during the knee extension exercise. Data from Steinkamp et al. show that patellofemoral stress progressively decreased as knee angle increased, peaking at 0° knee angle. In contrast, patellofemoral stress during the knee flexing phase in the current study progressively increased from approximately full knee extension to approximately 60° knee angle and then progressively decreased at higher knee angles as the knee continued flexing (Fig. 9). Similarly, patellofemoral stress during the knee extending phase progressively increased from approximately full knee flexion to approximately 60° knee angle and progressively decreased at lower knee angles as the knee continued extending. Since the patellofemoral compressive force curve from Kaufman et al. (26) had the same general shape and magnitude as that in the current study, it is deduced that patellofemoral stress data is similar in the study of Kaufman et al. and the current study. These patellofemoral stress data demonstrate that patellofemoral stress patterns differs between isometric knee extensions (55) and dynamic knee extensions (26). These findings are contrary to what many rehabilitation specialists believe concerning the knee extension exercise. It appears that the current thinking in many rehabilitation settings is that patellofemoral stress is highest at full knee extension, especially between 0–30° knee angle, which is in accord with isometric knee extension data from Steinkamp et al. (55). However, since patellofemoral data from both the current study and from Kaufman et al. (26) have implied that patellofemoral stress may be greater at higher knee angles (i.e., 60–70° knee angle) during a dynamic knee extension, the current views on patellofemoral stress and patellofemoral rehabilitation may need rethink-

ing. This is especially true since knee extension exercises are typically performed dynamically in rehabilitation settings, which is more functional compared with isometric contractions.

CONCLUSIONS

Judicious thought should be given in choosing exercises for rehabilitation or athletic training. Decisions should be made relative to which exercises best meet the intended goals of the rehabilitation or conditioning program. OKCE may be more effective in rectus femoris development, while CKCE may be more effective in developing the vastus medialis and vastus lateralis. Gastrocnemius development may be similar for all exercises, while the squats may be more effective in hamstring development. Since increased tibiofemoral compressive force has been shown to enhance knee stability by resisting anteroposterior translation, the higher compressive forces observed in OKCE at less than 30° knee angle and in CKCE at greater than 70° knee angle may aid in minimizing tensile forces in the cruciate ligaments. In OKCE the ACL is under tension at less than 25° knee angle and increased tension in the PCL occurs at greater than 65° knee angle in CKCE. Consequently, the higher compressive forces that occur during these knee flexion ranges may unload some of the tensile force in these respective cruciate ligaments. All exercises appear equally effective in minimizing ACL tensile force except the final 25° of knee extending in OKCE. Therefore, it may be prudent to exclude this range of motion for the patient using OKCE for rehabilitation after an ACL injury. OKCE is preferred over CKCE if minimal PCL tensile force is desired. Since PCL tension generally increased with knee flexion for all exercises, knee ranges of motion less than 60° knee angle will minimize PCL tensile force. After PCL injury, which typically occurs less often than ACL injuries, it may be prudent to limit knee flexion during exercise, especially at knee angles greater than 60°. Since patel-

lofemoral compressive force and stress increased in CKCE with knee flexion, those suffering with patellofemoral dysfunctions should employ low to mid-range knee angles (e.g., training within a more functional knee range between 0–50° knee angle) when training with CKCE. However, mid-range knee angles may exacerbate patellofemoral dysfunctions in OKCE, since peak patellofemoral stress was observed at approximately 60° knee angle (peak patellofemoral compressive force occurred at approximately 75° knee angle). Employing lower (e.g., 0–30° knee angle) or higher (e.g., 75–90° knee angle) knee angles may be most effective in minimizing patellofemoral dysfunctions, although the 0–30° knee angle range is currently not recommended in rehabilitation settings. Further research is needed concerning patellofemoral compressive force and stress in OKCE, since current data is inconclusive and contrary results have been reported.

To estimate the actual articulating tibiofemoral and patellofemoral compressive forces generated about the knee, muscle and ligamentous structures must be included in a biomechanics knee model that calculates muscle and ligamentous forces. Unfortunately, numerous assumptions are needed which may adversely affect the accuracy of these calculations. Additional studies are needed to corroborate these results, and continued improvements are needed in biomechanics knee models to increase the accuracy in calculating knee joint kinetics.

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